databricks trans / 2.driver_standings

(https://databricks.com)

```
dbutils.widgets.text("p_file_date", "2021-03-28")
v_file_date = dbutils.widgets.get("p_file_date")
```

```
%run "../includes/common_functions"
```

```
%run "../includes/configuration"
```

```
# race_results_df = spark.read.parquet(f"{presentation_folder_path}/race_results")
```

```
race_results_df = spark.read.format("delta").load(f"{presentation_folder_path}/race_results") \
.filter(f"file_date = '{v_file_date}'")
```

```
# display(race_results_df)
```

Table

	race_id	race_year	race_name	race_date	circuit_location	driver_name
1	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Mick Schumacher
2	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Nikita Mazepin
3	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Yuki Tsunoda
4	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Nicholas Latifi
5	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	George Russell
6	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Lando Norris
7	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Charles Leclerc
8	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Pierre Gasly
9	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Antonio Giovinazzi
10	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Lance Stroll
11	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Esteban Ocon
12	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Carlos Sainz
13	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Max Verstappen
14	1053	2021	Emilia Romagna Grand P...	2021-04-18T13:00:00.000+00:...	Imola	Valtteri Bottas

20 rows

```
race_year_list = df_column_to_list(race_results_df, 'race_year')
```

```
from pyspark.sql.functions import col

race_results_df = spark.read.format("delta").load(f"{presentation_folder_path}/race_results") \
.filter(col("race_year").isin(race_year_list))
```

```
from pyspark.sql.functions import sum, when, count, col

# driver_standings_df = race_results_df \
# .groupBy("race_year", "driver_name", "driver_nationality", "team") \
# .agg(sum("points").alias("total_points"),
#      count(when(col("position") == 1, True)).alias("wins"))

driver_standings_df = race_results_df \
.groupBy("race_year", "driver_name", "driver_nationality") \
.agg(sum("points").alias("total_points"),
     count(when(col("position") == 1, True)).alias("wins"))
```

```
from pyspark.sql.window import Window
from pyspark.sql.functions import desc, rank, asc
```

```
driver_rank_spec = Window.partitionBy("race_year").orderBy(desc("total_points"), desc("wins"))
final_df = driver_standings_df.withColumn("rank", rank().over(driver_rank_spec))
```

```
# from pyspark.sql.functions import sum

# driver_standings_df = race_results_df \
#     .groupBy("race_year", "driver_name", "driver_nationality", "team") \
#     .agg(sum("points").alias("total_points"))
```

```
# display(driver_standings_df.filter("race_year = 2020"))
```

```
# display(driver_standings_df.filter("race_year = 2020"))
```

```
# from pyspark.sql.window import Window
# from pyspark.sql.functions import desc, rank, asc

# driver_rank_spec = Window.partitionBy("race_year").orderBy(desc("total_points"))
# final_df = driver_standings_df.withColumn("rank", rank().over(driver_rank_spec))
```

```
# display(final_df.filter("race_year = 2020"))
```

```
# final_df.write.mode("overwrite").format("parquet").saveAsTable("f1_presentation.driver_standings")
```

```
merge_condition = "tgt.driver_name = src.driver_name AND tgt.race_year = src.race_year"
merge_delta_data(final_df, 'f1_presentation', 'driver_standings', presentation_folder_path, merge_condition, 'race_year')
```

```
# %sql
# select * from f1_presentation.driver_standings;
```

Table

	race_year	driver_name	driver_nationality	total_points	wins	rank
1	1952	Alberto Ascari	Italian	53.5	6	1
2	1952	Nino Farina	Italian	27	0	2
3	1952	Piero Taruffi	Italian	22	1	3
4	1952	Mike Hawthorn	British	10	0	4
5	1952	Rudi Fischer	Swiss	10	0	4
6	1952	Robert Manzon	French	9	0	6
7	1952	Troy Ruttman	American	8	1	7
8	1952	Luigi Villoresi	Italian	8	0	8
9	1952	José Froilán González	Argentine	6.5	0	9
10	1952	Jean Behra	French	6	0	10
11	1952	Jim Rathmann	American	6	0	10
12	1952	Sam Hanks	American	4	0	12
13	1952	Dennis Poore	British	3	0	13
14	1952	Ken Wharton	British	3	0	13
15	1952	Duane Carter	American	3	0	13

3,142 rows

```
# %sql
# SELECT * FROM f1_presentation.driver_standings WHERE race_year = 2021;
```

Table

	race_year	driver_name	driver_nationality	total_points	wins	rank
1	2021	Lewis Hamilton	British	44	1	1
2	2021	Max Verstappen	Dutch	43	1	2
3	2021	Lando Norris	British	27	0	3
4	2021	Charles Leclerc	Monegasque	20	0	4
5	2021	Valtteri Bottas	Finnish	16	0	5
6	2021	Daniel Ricciardo	Australian	14	0	6
7	2021	Carlos Sainz	Spanish	14	0	6
8	2021	Sergio Pérez	Mexican	10	0	8
9	2021	Pierre Gasly	French	6	0	9
10	2021	Lance Stroll	Canadian	5	0	10
11	2021	Esteban Ocon	French	2	0	11
12	2021	Yuki Tsunoda	Japanese	2	0	11
13	2021	Fernando Alonso	Spanish	1	0	13
14	2021	George Russell	British	0	0	14
15	2021	Antonio Giovinazzi	Italian	0	0	14

20 rows